

A Hierarchical Model for Longitudinal and Intraparticle Diffusion Coefficients in Liquid Chromatography

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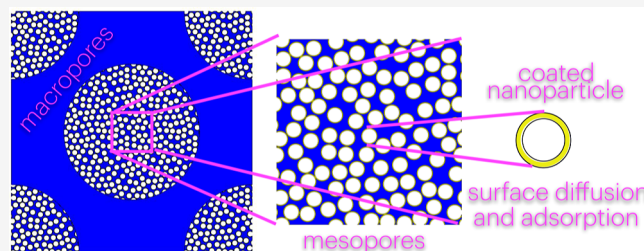
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ABSTRACT: The model for the effective longitudinal diffusion coefficient D_{eff} recently derived from the two-zone moment analysis approach, is adapted to predict the intraparticle diffusion coefficient D_{part} of the general rate model for chromatography. The model for D_{part} , coupled with the model for D_{eff} , allows us to predict with great accuracy numerical data for D_{eff} obtained by solving the detailed diffusion-adsorption transport model, via the Brenner–Adler method of moments, in different 2D and 3D hierarchical retentive porous structures. The proposed models make explicit the dependence of D_{eff} and D_{part} on the obstruction factors of the macroporous and mesoporous zone, internal and external porosity, equilibrium constant, specific surface area, and surface diffusion coefficient. Detailed numerical simulations also allowed us to verify the inaccuracy of the parallel-connection model for D_{part} when surface diffusion in the stationary phase layer contributes significantly to the transport of the analyte in the mesoporous zone.



INTRODUCTION

The peak-parking technique^{1–4} offers a way to experimentally measure the effective longitudinal diffusion coefficient D_{eff} and determine the B-term contribution^{5,6} which, in most columns, constitutes a significant fraction (order of 30–40%) of the plate height at the minimum of the van Deemter curve. This method has become a trusted tool in HPLC column characterization, helping to assess factors such as the structure of the stationary phase, surface chemistry, and retention mechanisms. D_{eff} values can also be examined for the determination of the intraparticle diffusivity D_{part} in the mesoporous region of the chromatographic bed. D_{part} is a crucial parameter in a column's efficiency. It determines the B-term dispersion and the intraparticle mass transfer resistance and also contributes significantly to the eddy-dispersion.⁷ D_{part} in turn is influenced by the obstruction effect of the intraparticle mesopore space, the actual composition of the mobile phase along the coordinate perpendicular to the mesopore walls,^{8,9} and the diffusion rate experienced in (partition-based chromatography) or on the stationary phase (adsorption-based chromatography).¹⁰ In the past, several models have been proposed to describe this, some based on the sum of parallel diffusion fluxes,¹ some based on the effective medium theory,⁶ and some including two-zone¹¹ and even three- and four-zone models.¹² D_{part} is a key parameter of the general rate (GR) model of chromatography^{13,14} which in the purely diffusive case (zero eluent velocity) reads as

$$\begin{aligned} \frac{\partial c_m}{\partial t} &= D_m \nabla^2 c_m, \quad \mathbf{x} \in \Omega_m \\ \frac{\partial c_{\text{part}}}{\partial t} &= D_{\text{part}} \nabla^2 c_{\text{part}}, \quad \mathbf{x} \in \Omega_{\text{part}} \\ c_{\text{part}}|_{\partial\Omega_m} &= K_{\text{part}} c_m|_{\partial\Omega_m} \\ -D_m \nabla c_m \cdot \mathbf{n}|_{\partial\Omega_m} &= -D_{\text{part}} \nabla c_{\text{part}} \cdot \mathbf{n}|_{\partial\Omega_m} \end{aligned} \quad (1)$$

where $\mathbf{n}$ is the unit norm vector at the fluid/particle interface $\partial\Omega_m$, c_m [mol/m³] and c_{part} [mol/m³] are analyte concentrations in the mobile zone Ω_m (volume V_{Ω_m}) and in the stationary/particle zone Ω_{part} (total volume $V_{\Omega_{\text{part}}}$ including mesopores) shown in Figure 1A, D_m and D_{part} the diffusion coefficients in Ω_m and Ω_{part} , and K_{part} the partition coefficient relating c_m and c_{part} at the boundary $\partial\Omega_m$

$$K_{\text{part}} = \varepsilon_{\text{int}}(1 + K_{\text{eq}}a) \quad (2)$$

and accounting for the mesoporous zone characteristic parameters, namely, the internal porosity

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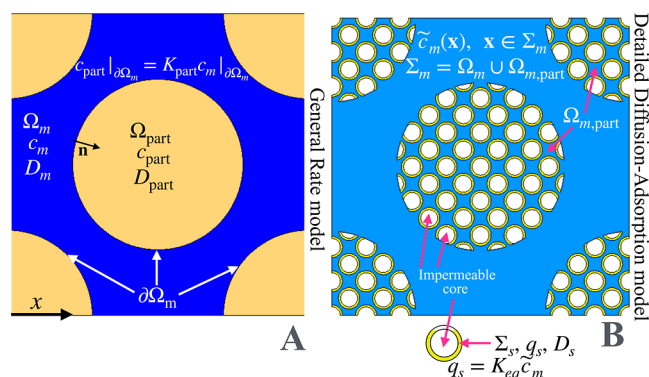


Figure 1. Domains, variables, and parameters entering the general rate model (A) and the detailed diffusion-adsorption model (B). The blue region in figure A indicates Ω_m . The azure region in figure B indicates Σ_m .

$$\varepsilon_{\text{int}} = \frac{\text{volume of the mesopores}}{\text{total particle volume}} = \frac{V_{\Omega_{m,part}}}{V_{\Omega_{part}}}, \text{ the specific surface}$$

$$a[\text{m}^{-1}] = \frac{\text{surface of the mesopores}}{\text{volume of the mesopores}} = \frac{S}{V_{\Omega_{m,part}}}, \text{ and the equilibrium}$$

constant $K_{eq}[\text{m}] = \frac{k_a[\text{ms}^{-1}]}{k_d[\text{s}^{-1}]}$, i.e., the ratio between the adsorption k_a and desorption k_b rate constants.

Being derived from a volume-averaging approach,¹⁵ the GR model considers the mesoporous region Ω_{part} as an isotropic and homogeneous solid phase, in which analyte transport occurs solely by diffusion with an effective diffusion coefficient D_{part} accounting for both hindered diffusion and surface adsorption in the mesopores, i.e., $D_{part} = f(\varepsilon_{\text{int}}\gamma_{m,int}^0 K_{eq}a, D_s)$, where $\gamma_{m,int}^0$ is the obstruction factor experienced by the analyte when diffusing in the particle mesopores $\Omega_{m,part}$ and D_s is the surface diffusion coefficient, i.e., the analyte diffusivity in the stationary phase layer covering the walls of the mesopores. When experimental data of D_{eff} are available, it is possible to follow a top-down approach, which consists in adopting a model for D_{eff} , for example, Maxwell's or Torquato's model^{6,16,17} derived from the effective medium theory (EMT) or the model recently proposed by Adrover et al.^{18,19} based on the two-zone moment approach (TZMA).²⁰ The model for D_{eff} can be used to estimate D_{part} as a best-fit parameter, for different column types and different retention conditions, thus investigating how D_{part} is affected by the mesopore geometry and physical parameters. When adopting a computational approach,^{21–24} a bottom-up strategy can be followed, which consists in solving a detailed transport model of the analyte in the macro-mesoporous chromatographic structure. In this way, it is possible to control the geometry of the mesoporous zone ($\varepsilon_{\text{int}}\gamma_{m,int}^0$), the retentive conditions ($K_{eq}a$), and the surface diffusion (D_s),²⁵ thus analyzing how all these mesoporous-scale parameters influence the intraparticle diffusivity D_{part} which, in turn, controls the macroscopic effective longitudinal diffusion coefficient D_{eff} . Computationally, this is only possible if one adopts a method that allows estimation of D_{eff} by solving the transport equations of the detailed model only on a unit cell representative of the entire chromatography bed. In this paper, the Brenner–Adler method of moments²⁶ is adopted to estimate D_{eff} from spatial moments of the analyte concentrations $\tilde{c}_m$ and q_s satisfying the detailed diffusion-adsorption (DDA) model^{26,27} in the limiting

case of negligible mass transfer resistance at the fluid/solid interface and infinitely fast adsorption/desorption process

$$\begin{aligned} \frac{\partial \tilde{c}_m}{\partial t} &= D_m \nabla^2 \tilde{c}_m, \quad \mathbf{x} \in \Sigma_m = \Omega_m \cup \Omega_{m,part} \\ \frac{\partial q_s}{\partial t} &= D_s \nabla_s^2 q_s - D_m \nabla \tilde{c}_m \cdot \mathbf{n}|_{\Sigma_s}, \quad \mathbf{x} \in \Sigma_s \\ q_s|_{\Sigma_s} &= K_{eq} \tilde{c}_m|_{\Sigma_s} \end{aligned} \quad (3)$$

In this setting, $\tilde{c}_m$ [mol/m³] is the analyte concentration in the mobile zone Σ_m which includes both macropores Ω_m and mesopores $\Omega_{m,part}$ thus extending over a total volume $V_{\Sigma_m} = V_{\Omega_m} + V_{\Omega_{m,part}}$, where $V_{\Omega_{m,part}}$ is the fluid volume within the mesoporous particle (see Figure 1B). Correspondingly, q_s [mol/m²] is the analyte concentration in the stationary phase layer Σ_s (total surface area S) covering the walls of the mesopores (see Figure 1B), where the analyte is adsorbed and diffuses²⁸ with diffusion coefficient D_s . By adopting the Brenner–Adler method of moments, we analyzed and quantified the influence of retentive conditions and surface diffusion on D_{eff} for different geometries and internal porosities of the mesoporous zone. This allowed us to verify the reliability and accuracy of a model for D_{part} which was used in conjunction with a previously developed model for D_{eff} which is able to predict the numerical data of D_{eff} for the different geometries investigated and throughout the wide range of values of $K_{eq}a$ and D_s analyzed. The approach here proposed to estimate D_{part} and D_{eff} is not as accurate and detailed as that recently implemented by Tallarek et al.²⁹ combining molecular dynamics simulations in a single-mesopore model with Brownian dynamics simulations in hierarchical porosity models based on physical reconstructions of monolithic silica columns. However, the estimate of D_{eff} via the Brenner–Adler moment method in relatively simple hierarchical macropore–mesopore models permits us to assess the validity of a general model for D_{part} that overcomes the intrinsic limitations of the following oversimplified model

$$\frac{D_{part}^{\parallel}}{D_m} = \frac{\gamma_{m,int}^0}{1 + K_{eq}a} + \frac{K_{eq}a}{1 + K_{eq}a} \frac{D_s}{D_m} \quad (4)$$

derived from the parallel-connection assumption that surface and pore diffusion occur in parallel.^{6,30}

The parallel-connection model, eq 4, is based on the idea of representing the mesoporous zone as a bundle of cylindrical mesopores arranged in parallel, although with a certain degree of tortuosity. However, a more realistic depiction of the mesoporous zone is as a sintered network of nanoscale spheres—a ternary system composed of the mobile phase liquid occupying the mesopores, impermeable silica particles, and a stationary phase layer coating the silica surface. This latter is the idea of the mesoporous zone that inspired the 2D and 3D model geometries investigated.

MATERIALS AND METHODS

According to the homogenization method proposed by Brenner and Alder,²⁶ the effective longitudinal diffusion coefficient D_{eff} of an analyte satisfying the transport eq 3 of the DDA model can be evaluated as

$$D_{\text{eff}} = D_m m_{\Sigma_m} (1 - \langle \nabla b_m \cdot \mathbf{e}_x \rangle_{\Sigma_m}) + D_s m_{\Sigma_s} (1 - \langle \nabla b_s \cdot \mathbf{e}_x \rangle_{\Sigma_s}) \quad (5)$$

where m_{Σ_m} and $m_{\Sigma_s} = 1 - m_{\Sigma_m}$ are the fractions of analyte particles in the mobile zone Σ_m and stationary phase layer Σ_s , respectively

$$m_{\Sigma_m} = \left(1 + \frac{K_{eq} S}{V_{\Sigma_m}} \right)^{-1} = \left(1 + K_{eq} a \frac{V_{\Omega_{m,part}}}{V_{\Sigma_m}} \right)^{-1} \\ = \left(1 + K_{eq} a \frac{(1 - \varepsilon_e) \varepsilon_{int}}{\varepsilon_{tot}} \right)^{-1} \quad (6)$$

$\varepsilon_e = V_{\Omega_m} / (V_{\Omega_m} + V_{\Omega_{part}})$ is the external porosity of the bed, $\varepsilon_{tot} = V_{\Sigma_m} / (V_{\Omega_m} + V_{\Omega_{part}}) = \varepsilon_e + (1 - \varepsilon_e) \varepsilon_{int}$ is the total porosity of the bed, and b_m [m] and b_s [m] are the b-fields satisfying the following stationary transport equations and boundary conditions on the unit cell $\Sigma = \Sigma_m \cup \Sigma_s$

$$D_m \nabla^2 b_m = 0, \quad \mathbf{x} \in \Sigma_m \\ K_{eq} D_s \nabla^2 b_s = -D_m (\nabla b_m - \mathbf{e}_x) \cdot \mathbf{n}|_{\Sigma_s} \\ b_m|_{\Sigma_s} = b_s|_{\Sigma_s} \quad (7)$$

plus periodic boundary conditions at the edges of the unit cell Σ . The vector $\mathbf{e}_x$ in eq 7 is the unit vector along the longitudinal direction x .

Figure 2A–C shows the 2D unit cells Σ ($\varepsilon_e = 0.5$) characterized by three different geometries of the mesoporous

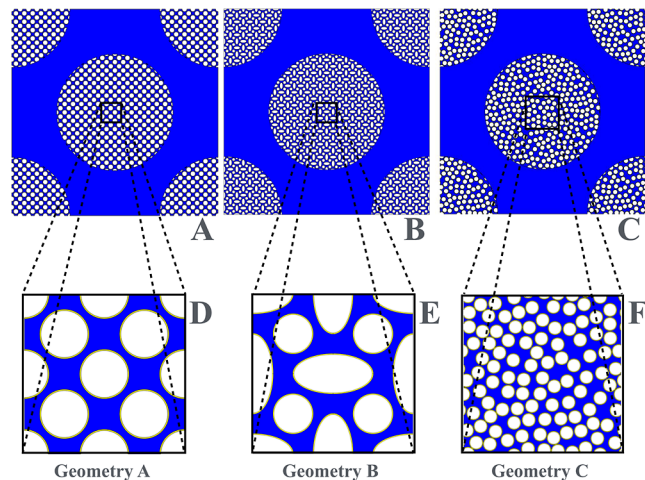


Figure 2. 2D unit cells with different geometries of the mesoporous zone. (A) 2D array of circles; (B) 2D array of circles and ellipses; and (C) 2D random arrangement of circles. The three insets (D–F) show the periodic unit cells of the mesoporous zone.

zone investigated, namely, a square array of circles (geometry A), an array of circles and ellipses (geometry B), and a random arrangement of circles (geometry C). Figure 3B shows the 3D unit cell of geometry D (simple cubic array of spheres, $\varepsilon_e \approx 0.49$) and the geometry of the mesoporous zone made by 100 equal spheres packed in a sphere³¹ ($\varepsilon_{int} \approx 0.48$). The coupled transport eq 7 for the b-fields has been solved by Finite Element Approach (FEM) in Comsol Multiphysics 6.3. The coefficient form partial differential equation (PDE) physics coupled with the coefficient form boundary PDE physics have

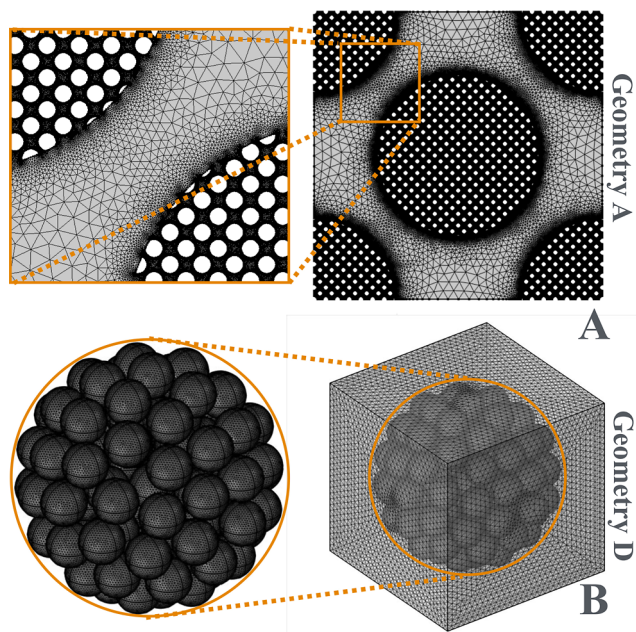


Figure 3. Mesh adopted for the 2D and 3D periodic unit cells. (A) 2D geometry A and (B) 3D geometry D.

been utilized with Lagrangian quadratic shape functions. The MULTifrontal Massively Parallel Sparse direct Solver (MUMPS) linear solver was employed with a relative tolerance of 1×10^{-6} . The triangular mesh of 2D geometries (Figure 3A) consists of approximately 7×10^5 domain elements, 1×10^5 boundary elements (see Figure 3A). The tetrahedral mesh of the 3D geometry (Figure 3B) consists of approximately 3×10^6 domain elements, 4×10^5 boundary elements, and 3×10^4 edge elements (see Figure 3B). Results are verified to be mesh independent in the whole range of $K_{eq}a$ and D_s/D_m values investigated.

RESULTS AND DISCUSSION

Numerical results for the rescaled effective longitudinal diffusion coefficient $\gamma_{\text{eff}} = D_{\text{eff}}/D_m$ vs $K_{eq}a$ for different values of the rescaled surface diffusion coefficient $D_s/D_m \in [0, 0.5]$ are shown in Figure 4 for geometry A with $\varepsilon_e = 0.5$ and $\varepsilon_{int} = 0.4$, corresponding to a total porosity of $\varepsilon_{tot} = 0.7$. It can be observed that all the curves satisfy the limiting behavior $\lim_{K_{eq}a \rightarrow 0} \gamma_{\text{eff}} = \gamma_{\Sigma_m}^0$, where $\gamma_{\Sigma_m}^0$ is the obstruction factor of Σ_m . Moreover, $\lim_{K_{eq}a \rightarrow \infty} \gamma_{\text{eff}} = 0$ because the mesoporous zone is made by isolated particles, and the chromatographic bed is not a bicontinuous structure. The higher the surface diffusivity D_s , the higher γ_{eff} . For very low values of D_s/D_m , namely, $D_s/D_m \leq 10^{-2}$, curves deviate only for very high values of $K_{eq}a$ from the limit curve with zero surface diffusion

$$\gamma_{\text{eff}}|_{D_s=0} = m_{\Sigma_m} \gamma_{\Sigma_m}^0 \quad (8)$$

while they are extremely sensitive to D_s for $D_s/D_m > 0.1$ in a wide range of $K_{eq}a$ values. γ_{eff} is rather insensitive to the geometry of the mesoporous phase, as shown in Figure 5A,B where the curves γ_{eff} vs $K_{eq}a$ are shown for geometry A in comparison with geometry B (Figure 5A) and geometry C (Figure 5B) for the same external porosity $\varepsilon_e = 0.5$ and the same internal porosity $\varepsilon_{int} = 0.4$. Small differences between geometry A and geometry C can be observed for very low $K_{eq}a$

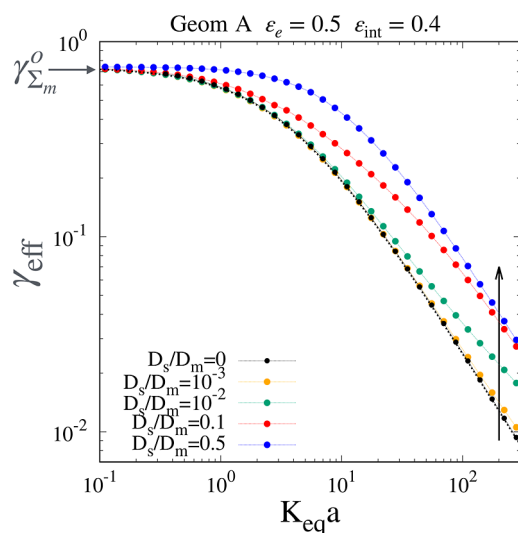


Figure 4. Numerical results for γ_{eff} vs $K_{\text{eq}}a$ for geometry A with $\varepsilon_e = 0.5$ and $\varepsilon_{\text{int}} = 0.4$. The arrow indicates increasing values of D_s/D_m . The dotted black line shows eq 8 with $\gamma_{\Sigma_m}^0 = 0.752$.

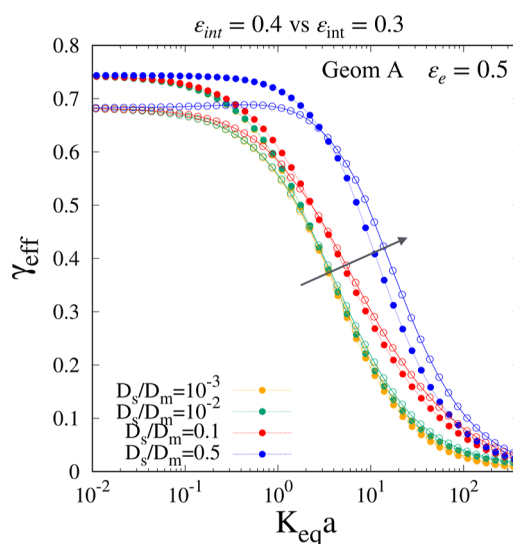


Figure 6. γ_{eff} vs $K_{\text{eq}}a$ for $\varepsilon_e = 0.5$ for geometry A with $\varepsilon_{\text{int}} = 0.4$ (full points) and $\varepsilon_{\text{int}} = 0.3$ (open circles). The arrow indicates increasing values of D_s/D_m .

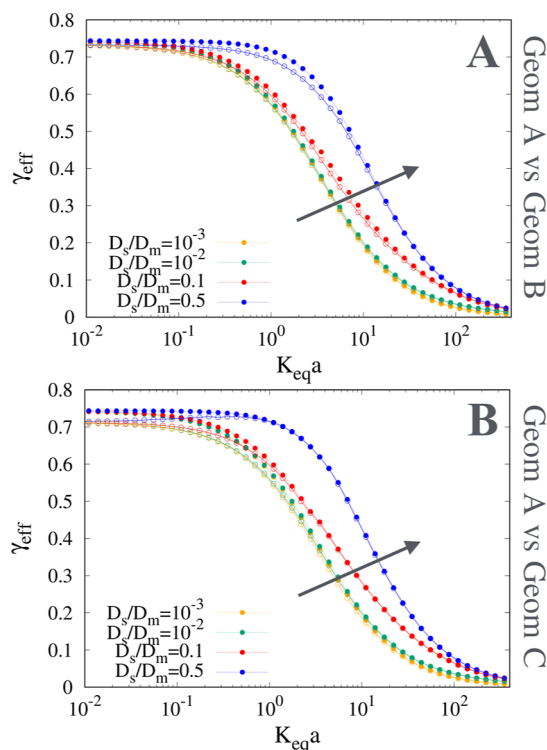


Figure 5. γ_{eff} vs $K_{\text{eq}}a$ for $\varepsilon_e = 0.5$, $\varepsilon_{\text{int}} = 0.4$ and for different values of D_s/D_m . (A) Geometry A (full points) vs geometry B (open circles). (B) Geometry A (full points) vs geometry C (open circles). Arrows indicate increasing values of D_s/D_m .

values because the obstruction factor $\gamma_{\Sigma_s}^0$ for random circles (geometry C) is appreciably smaller than that for the ordered array of circles (geometry A). On the contrary, γ_{eff} is significantly affected by the internal porosity ε_{int} because this latter strongly influences the obstruction factor $\gamma_{\Sigma_m}^0$ and the behavior of the γ_{eff} vs $K_{\text{eq}}a$ for $D_s/D_m \geq 0.1$, as shown in Figure 6, for geometry A and for two different internal porosities, namely, $\varepsilon_{\text{int}} = 0.3, 0.4$.

Modeling D_{eff} and D_{part} . Recently, a general model for D_{eff} , derived from the TZMA approach, has been proposed as a valid alternative to the EMT models used so far in the literature. Its advantage is that it does not require any fitting parameter because all the necessary information is embedded into the experimentally accessible geometric obstruction factor $\gamma_{\Omega_m}^0$ of the mobile phase Ω_m , originating from the tortuosity of the through-pore space (limiting case of fully solid particles without any retention). In its more general version, reliable for diluted and concentrated packings, the model reads as

$$\gamma_{\text{eff}} = \frac{\varepsilon_e \gamma_{\Omega_m}^0 + (1 - \varepsilon_e \gamma_{\Omega_m}^0 + \beta - \delta) \alpha_{\text{part}} + \delta \alpha_{\text{part}}^{1-\varepsilon_e}}{\varepsilon_e (1 + k'') (1 + (\beta - \delta) \alpha_{\text{part}} + \delta \alpha_{\text{part}}^{1-\varepsilon_e})} \quad (9)$$

where α_{part} is the relative particle permeability⁶ defined with respect to the zone retention factor k''

$$\alpha_{\text{part}} = k'' \frac{\varepsilon_e}{(1 - \varepsilon_e)} \frac{D_{\text{part}}}{D_m}$$

$$k'' = K_{\text{part}} \frac{V_{\Omega_{\text{part}}}}{V_{\Omega_m}} = \varepsilon_{\text{int}} (1 + K_{\text{eq}}a) \frac{1 - \varepsilon_e}{\varepsilon_e} \quad (10)$$

and β and δ are geometric parameters that depend solely on ε_e and $\gamma_{\Omega_m}^0$ as follows

$$\beta = \frac{\varepsilon_e (1 - \gamma_{\Omega_m}^0)}{(1 - \varepsilon_e)}, \quad \delta = \frac{\left(\frac{n}{n+1-\varepsilon_e} - \gamma_{\Omega_m}^0 \right)}{(1 - \varepsilon_e \gamma_{\Omega_m}^0)} \quad (11)$$

where $n = 1, 2$ for cylindrical and spherical particles, respectively. In the diluted case ($\varepsilon_e \geq 0.5$), the model attains the simplified form

$$\gamma_{\text{eff}} = \frac{\varepsilon_e \gamma_{\Omega_m}^0 + (1 - \varepsilon_e \gamma_{\Omega_m}^0 + \beta) \alpha_{\text{part}}}{\varepsilon_e (1 + k'') (1 + \beta \alpha_{\text{part}})} \quad (12)$$

because $\gamma_{\Omega_m}^0 \simeq \frac{n}{n+1-\varepsilon_e}$ (Maxwell obstruction factor for diluted suspensions), that implies $\delta = 0$.

The model applies to both fully porous and shell-porous particles by expressing D_{part} and k'' as

$$D_{\text{part}} = D_{\text{pz}} \left(\frac{n}{n + \rho^{n+1}} \right), \quad k'' = K_{\text{pz}} \frac{V_{\Omega_{\text{pz}}}}{V_{\Omega_m}} \quad (13)$$

where $\rho = d_{\text{core}}/d_{\text{part}}$ is the relative diameter of the solid particle core. D_{pz} and K_{pz} are the diffusion coefficient and the partition coefficient in the mesoporous zone Ω_{pz} that can be the entire particle if $\rho = 0$ or only the shell layer for $\rho > 0$. In the case of fully porous particles, $\Omega_{\text{part}} = \Omega_{\text{pz}}$, $D_{\text{part}} = D_{\text{pz}}$, and $K_{\text{part}} = K_{\text{pz}}$. To compute D_{pz} needed in eq 13, we should zoom in on the mesopore level. However, at this level, we again see a ternary medium (mobile phase liquid occupying the mesopores and impermeable nanoparticles coated with a stationary phase layer cf. Figure 2) that should obey the same phenomenological laws for the effective diffusion as described by eqs 9–13 for the whole-column level (mobile phase liquid occupying the macropores of a packing of shell-porous particles). Assuming the mesopore space consists of a packing of surface-coated nanoparticles (cf. Figure 2) with surface diffusion D_s , the translation of eqs 9–13 to the mesopore space requires that (i) γ_{eff} in eq 9 is replaced by D_{pz}/D_m . (ii) The mobile phase Ω_m is replaced by the mobile phase internal to the mesoporous particle $\Omega_{m,\text{part}}$, thus replacing ε_e with ε_{int} and $\gamma_{\Omega_m}^0$ with $\gamma_{\Omega_{m,\text{part}}}^0$. (iii) The zone retention factor k'' is replaced with $K_{\text{eq}}a$ because K_{pz} must be replaced by K_{eq} and $V_{\Omega_{\text{pz}}}/V_{\Omega_m}$ with $a = S/V_{\Omega_{m,\text{part}}}$. (iv) D_{part} is replaced by $\frac{D_s}{D_m} \frac{n}{n+1}$ by adopting the Hashin–Shtrikman formulation⁶ in eq 13 with the assumption that the coating (with internal diffusion D_s) on the nanoparticles is so thin that we can put $\rho = 1$ in eq 13.

The resulting expression of D_{pz}/D_m reads as

$$\frac{D_{\text{pz}}}{D_m} = \frac{\varepsilon_{\text{int}} \gamma_{m,\text{int}}^0 + (1 - \varepsilon_{\text{int}} \gamma_{m,\text{int}}^0 + \beta_{\text{int}} - \delta_{\text{int}}) \alpha_{\text{int}} + \delta_{\text{int}} \alpha_{\text{int}}^{1-\varepsilon_{\text{int}}}}{\varepsilon_{\text{int}} (1 + K_{\text{eq}}a) (1 + (\beta_{\text{int}} - \delta_{\text{int}}) \alpha_{\text{int}} + \delta_{\text{int}} \alpha_{\text{int}}^{1-\varepsilon_{\text{int}}})} \quad (14)$$

$$\beta_{\text{int}} = \frac{\varepsilon_{\text{int}} (1 - \gamma_{m,\text{int}}^0)}{(1 - \varepsilon_{\text{int}})}, \quad \delta_{\text{int}} = \frac{\left(\frac{n}{n+1-\varepsilon_{\text{int}}} - \gamma_{m,\text{int}}^0 \right)}{(1 - \varepsilon_{\text{int}} \gamma_{m,\text{int}}^0)} \quad (15)$$

$$\alpha_{\text{int}} = K_{\text{eq}}a \frac{\varepsilon_{\text{int}}}{1 - \varepsilon_{\text{int}}} \frac{D_s}{D_m} \frac{n}{n+1} \quad (16)$$

Comparison with Numerical Results. Figure 7A,B shows the excellent agreement between model predictions based on eqs 14–16 and numerical results for D_{pz}/D_m vs $K_{\text{eq}}a$ obtained by solving the b-field equations, eq 7, within different 2D and 3D periodic unit cells representative of the mesoporous zone, namely, the 2-d array of circles (Geom A, Figure 2D), the 2D random arrangement of circles (Geom C, Figure 2F), the 3D face-centered cubic array of spheres (fcc), and a 3D close random packing (CRP) of 100 equal spheres ($\varepsilon_{\text{int}} = 0.375$) with periodic boundary conditions generated using a collision-driven packing algorithm.³² The values of the mesopore obstruction factor $\gamma_{m,\text{int}}^0$ for the different mesopore unit cells investigated, are reported in Table 1. Indeed, $\gamma_{m,\text{int}}^0$ is the unique parameter entering the D_{pz} model, eq 14, and is

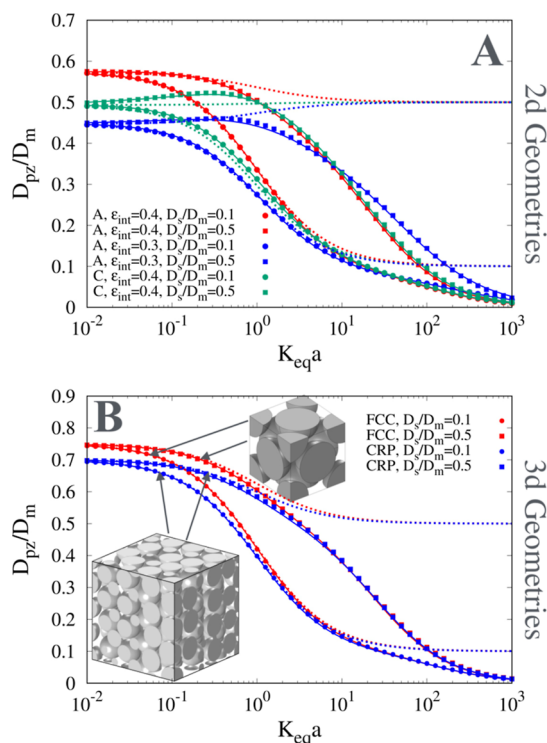


Figure 7. Comparison between numerical results (points) and model predictions for D_{pz}/D_m vs $K_{\text{eq}}a$ for $D_s/D_m = 0.1, 0.5$. Continuous lines: TZMA model, eq 14. Dashed lines: parallel-connection model, eq 4. (A) 2D geometry A ($\varepsilon_{\text{int}} = 0.3, 0.4$) and geometry C ($\varepsilon_{\text{int}} = 0.4$). (B) 3D geometries: fcc array of spheres (fcc, $\varepsilon_{\text{int}} = 0.4$) and a close random packing of 100 equal spheres (CRP, $\varepsilon_{\text{int}} = 0.375$).

Table 1. $\gamma_{m,\text{int}}^0$ for Different Mesoporous Geometries^a

	geom A	geom B	geom C
$\gamma_{m,\text{int}}^0$	0.575 [0.4]	0.545 [0.4]	0.505 [0.4]
	geom A	fcc	CRP
$\gamma_{m,\text{int}}^0$	0.448 [0.3]	0.750 [0.4]	0.703 [0.375]

^aThe corresponding porosity ε_{int} is indicated in square brackets [·].

obtained by solving the b_m -field equation in the mesopores $\Omega_{m,\text{part}}$ in the impermeable case, i.e., in the limiting case of no superficial adsorption, $K_{\text{eq}}a = 0$. Figure 7A,B also shows that the parallel-connection model, eq 4, significantly overestimates D_{pz} especially for intermediate/large values of D_s/D_m .

By replacing eq 14 for $D_{\text{pz}}(\varepsilon_{\text{int}} \gamma_{m,\text{int}}^0 K_{\text{eq}}a, D_s)$ into eqs 9–13 for $\gamma_{\text{eff}} = f(\varepsilon_e, \gamma_{\Omega_m}^0, K_{\text{eq}}a, D_{\text{part}})$, we obtain a complete model for γ_{eff} that makes explicit the dependence on the internal and external porosities, the equilibrium constant, the specific surface area, and surface diffusion coefficient. The dependence on the specific geometries of the macropore and mesopore networks is embedded in the obstruction factors $\gamma_{\Omega_m}^0$ and $\gamma_{m,\text{int}}^0$. For all the 2D geometries investigated, the macropore structure is the same (square array of circles, $\varepsilon_e = 0.5$) and the corresponding macropore obstruction factor is $\gamma_{\Omega_m}^0 = 0.652$. For the 3D geometry D, the macropore structure is a simple cubic array of spheres ($\varepsilon_e = 0.49$), and the corresponding obstruction factor is $\gamma_{\Omega_m}^0 = 0.789 \approx 2/(3 - \varepsilon_e)$. Regarding the obstruction factor $\gamma_{m,\text{int}}^0$ in the mesoporous zone ($\varepsilon_{\text{int}} \approx 0.48$), since the 3D geometry D does not allow us to identify a

periodic cell, and thus to estimate $\gamma_{m,int}^0$, we assumed $\gamma_{m,int}^0 = \gamma_{\Omega_m}^0$ as the internal and external porosities are very similar and in the range of validity of the Maxwell equation. For all the geometries, $D_{part} = D_{pz}$ because only fully porous particles are considered (cf. Figures 2 and 3). Figure 8A shows

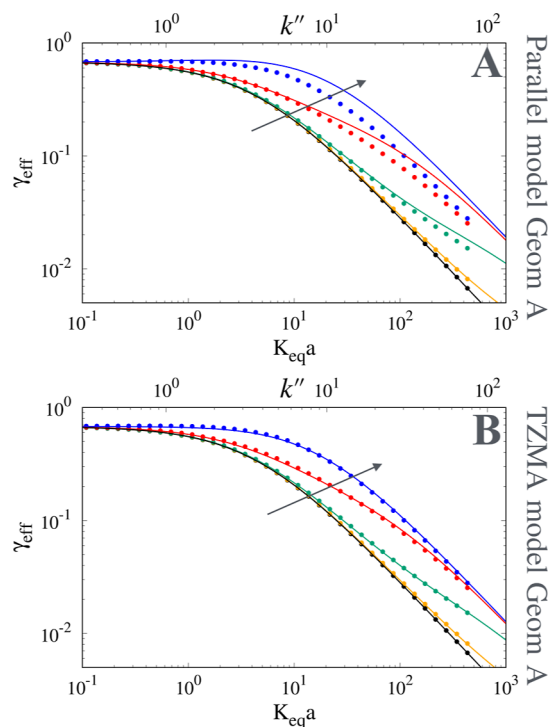


Figure 8. Comparison between numerical results (points) and model predictions (continuous lines) for γ_{eff} vs $K_{eq}a$ (bottom x axis) and k'' (top x axis) for geometry A, $\varepsilon_e = 0.5$, $\varepsilon_{int} = 0.3$. (A) Parallel-connection model, eq 4. (B) TZMA model, eq 14. The arrow indicates increasing values of $D_s/D_m = 0, 10^{-3}, 10^{-2}, 0.1$, and 0.5 .

the comparison, for geometry A, between numerical results for γ_{eff} vs $K_{eq}a$ (bottom x axis) and k'' (top x axis) and the predictions of the model for γ_{eff} eq 9 with $D_{pz} = D_{part}$ estimated from the parallel-connection model, eq 4. It can be observed that the adoption of $D_{part}^{||}$ results in a significant overestimate of γ_{eff} for $D_s/D_m \geq 0.1$ over the entire range of zone retention factor k'' of practical interest, $k'' \in [1, 30]$. In contrast, the adoption of the TZMA model for D_{pz} , eq 14, into the TZMA model for γ_{eff} , eq 9, leads to a very accurate prediction of numerical data for γ_{eff} in the whole range of D_s and k'' values investigated, as shown in Figure 8B for the 2D geometry A, and in Figure 9A–C for the 2D geometries B and C and for the 3D geometry D.

CONCLUSIONS

The model proposed for D_{eff} eq 9, and D_{pz} eq 14, allows us to predict with great accuracy (error below 2%) numerical data for 2D and 3D hierarchical porous media in the whole range of D_s/D_m and $K_{eq}a$ values investigated. The 2D geometries analyzed resemble closely those of pillar array columns with fully porous pillars. The mesoporous structures of the 3D geometries are reasonable models of monodisperse silica-based materials. Numerical results for D_{pz} and D_{eff} show low sensitivity to the specific geometry of the mesoporous zone and high sensitivity to the internal porosity. We therefore

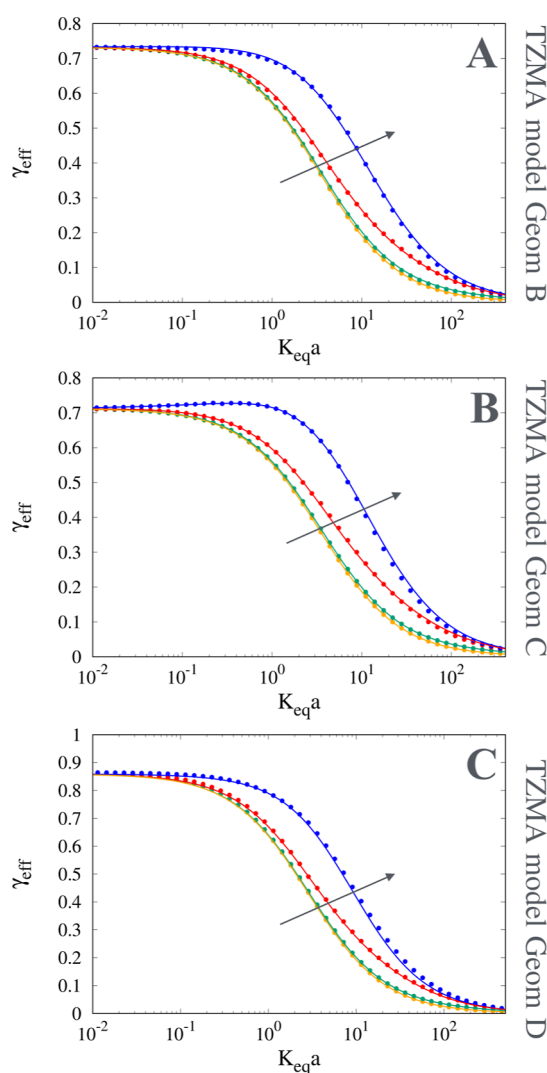


Figure 9. Comparison between numerical results (points) and model predictions (continuous lines) for γ_{eff} vs $K_{eq}a$. (A) 2D geometry B. (B) 2D geometry C. (C) 3D geometry D, $\varepsilon_e \approx 0.49$, $\varepsilon_{int} \approx 0.48$. The arrow indicates increasing values of $D_s/D_m = 10^{-3}, 10^{-2}, 0.1$, and 0.5 .

believe that, even though the geometries analyzed are simplified models of the complex structure of the macro-/mesoporous zone, the models proposed for D_{eff} and D_{pz} can also be successfully applied to more realistic geometries that can be obtained via porous-media reconstructions by scanning transmission electron microscopy (mesopore space)³³ and confocal laser scanning microscopy (macropore space).³⁴ The application of the hierarchical model $D_{pz}-D_{eff}$ requires the preliminary estimate of the two obstruction factors $\gamma_{\Omega_m}^0$ and $\gamma_{m,int}^0$. While $\gamma_{\Omega_m}^0$ can be experimentally estimated, $\gamma_{m,int}^0$ must be obtained by a best fit of experimental data. A good first attempt value of $\gamma_{m,int}^0$ is represented by the Maxwell diluted limit value $\gamma_{m,int}^0 \approx n/(n + 1 - \varepsilon_{int})$. Verification of the model is planned through application to a data set of experimental D_{eff} values obtained by Deridder et al.³⁵ using peak parking on various first- and second-generation monolithic silica columns. The hierarchical model proposed in this paper is derived from the model for D_{eff} stemming from the TZMA approach. By adopting a similar strategy, different hierarchical models can also be constructed from other models of D_{eff} that can be found in the literature, such as the

Torquato's model¹⁸ stemming from EMT, as already proposed by Desmet and Deridder back in 2011.⁶

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Notes

The authors declare no competing financial interest.

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ADDITIONAL NOTE

^aThe operators ∇_s and $\nabla_s^2 = \nabla_s \cdot (\nabla_s)$ are the surface gradient and Laplace–Beltrami operators, respectively. Specifically, $\nabla_s = \mathbf{I}_s \cdot \nabla$ wherein $\mathbf{I}_s = \mathbf{I} - \mathbf{n}\mathbf{n}$ is the surface idemfactor dyadic, $\mathbf{I}$ the idemfactor dyadic, and $\mathbf{n}$ a unit normal vector on Σ_s .

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